

EE 531: Estimation and Detection Theory

Course Code	EE 531
Course Name	Estimation and Detection Theory
L-T-P-C	3-0-0-3
Prerequisites	EE304/ EE503/ EE305/ MA524 or equivalent, or instructor consent
Intended for	B.Tech./M.S./M.Tech./Ph.D.
Distribution	Core for M.Tech.(CSP), Elective for B.Tech. 3rd and 4th year, M.S., M.Tech(non CSP), Ph.D.
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)

Course Contents

- **Mathematical background:** [3 Lectures]
 - Random variables and processes: vector spaces of random variables, CLT, stochastic processes, stationarity, spectral representation of stochastic processes, Wiener-Khinchin theorem, Gaussian process, white noise, stochastic processes through LTI systems.
 - Modeling of signals and noise: models, and selection of model and model order.
- **Ideal estimation for deterministic parameters:** [6 Lectures]
 - Principle of estimation and its applications, properties of estimators.
 - Minimum Variance Unbiased Estimation: existence and search of MVU estimators, sufficient statistics and its role in finding MVU estimator, extension to vector parameters. Neyman-Fisher factorization and Rao-Blackwell theorems.
 - Cramer-Rao Lower Bound: signals in white Gaussian noise. parameter transformation, vector parameters, general Gaussian case, and WSS Gaussian random processes, efficiency.
- **Practical estimation for deterministic parameters:** [5 Lectures]
 - Linear Models and Unbiased Estimators, scalar and vector Best Linear Unbiased Estimators (BLUE); Maximum Likelihood estimation, expectation-maximization (EM) algorithm; Least Squares estimation: linear, order-recursive, sequential, constrained, and non-linear.
- **Estimation for random parameters:** [6 Lectures]
 - Bayesian Estimation: Bayesian linear model, nuisance parameters. Bayesian estimation for deterministic parameters, risk functions, MMSE and MAP estimators, scalar, vector, and sequential Linear MMSE estimators, Wiener filtering.
 - Advanced topics: Levinson-Durbin and Innovation algorithms, graphical models, hidden Markov models.
- **Estimation beyond stationarity:** [4 Lectures]
 - Kalman filtering: State-space modeling, scalar, vector, and extended Kalman filters.
- **Statistical Detection Theory:** [5 Lectures]
 - Binary and multiple hypothesis testing, Neyman-Pearson theorem, receiver operating characteristics, minimum Bayes risk detectors, sequential detection, Composite hypothesis testing: Bayesian and generalized likelihood ratio test (GLRT), locally most powerful (LMP) detectors, asymptotically equivalent tests.
- **Detection of deterministic signals:** [4 Lectures]
 - Signals with known parameters: matched filter, linear model, multiple signal detection. Signals with unknown parameters: signal modeling and detector performance, sinusoidal detection, linear models, energy detectors.
- **Detection of random signals:** [5 Lectures]
 - Signals with known parameters: Estimator-correlator, linear model, general Gaussian detection.
 - Signals with unknown parameters: incompletely known signal covariance, weak signal detection.
- **Detection with non-Gaussian and colored noise:** [4 Lectures]
 - Signals with known and unknown parameters, Karhunen-Loeve expansion and whitening filters.
 - Advanced topics: Complex and vector extensions of detectors: known deterministic signal in CWGN, spatially/temporally uncorrelated noise, random signal in CWGN.

Text Books

1. S. M. Kay, **Fundamentals of Statistical Signal Processing**, Vol I: Estimation Theory, Prentice Hall, 1993.
2. S. M. Kay, **Fundamentals of Statistical Signal Processing**, Vol 2: Detection Theory, Prentice Hall, 1998.

References

1. L. L. Scharf, **Statistical Signal Processing: Detection, Estimation, and Time Series Analysis**, Addison-Wesley, 1991.
2. H. Y. Poor, **An Introduction to Signal Detection and Estimation**, Springer-Verlag, 1994.
3. C. W. Helstrom, **Elements of Signal Detection and Estimation**, Prentice Hall, 1995.
4. G. Casella and R. L. Berger, **Statistical Inference**, Duxbury Press, 2002.
5. H. L. van Trees, K. L. Bell, and Z. Tian, **Detection, Estimation, and Modulation Theory, Part I: Detection, Estimation, and Filtering Theory**, Wiley, 2013.